

Pepsi Cola Distribution (Riyadh) — Model Assumptions

Sales Assumptions			
Sales Volume (cans / month)	1,000,000		
Distributor Purchase Price (SAR / can)	1.00 SAR		
Selling Price to Retailers (SAR / can)	1.50 SAR		
Expected Returns / Waste (% of units)	1.0%		
Cost Assumptions			
Logistics & Transport (SAR / can)	0.10 SAR		
Warehouse Cost (SAR / month)	150,000 SAR		
Sales Team Salaries (SAR / month)	200,000 SAR		
Marketing Spend (SAR / month)	100,000 SAR		
Other Operating Expenses (SAR / month)	50,000 SAR		
Investment & Capital			
Initial Working Capital Needed (SAR)	2,000,000 SAR		
Equipment / Freezer CapEx (SAR)	500,000 SAR		
Depreciation Period (years, straight-line)	5		
Tax & Finance			
Corporate Tax Rate (%)	20.0%		
Discount Rate (%)	10.0%		
Project / Analysis Horizon (years)	5		

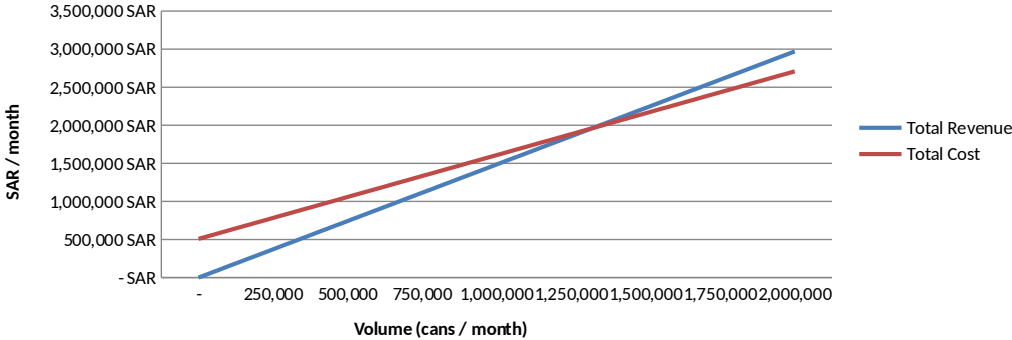
Yellow cells are editable inputs. All other cells are formulas that recalculate automatically.

Monthly P&L (steady-state month, all assumptions)	
Volume & Revenue	
Units Purchased (cans)	1,000,000
Returns / Waste %	1.0%
Units Wasted (cans)	10,000
Units Sold (cans)	990,000
Selling Price (SAR / can)	1.50 SAR
Revenue	1,485,000 SAR
Cost of Goods Sold	
Purchase Price (SAR / can)	1.00 SAR
COGS (all units purchased)	1,000,000 SAR
Logistics Cost (SAR / can)	0.10 SAR
Total Logistics & Transport Cost	100,000 SAR
Gross Margin	
Gross Profit	385,000 SAR
Gross Margin %	25.9%
Operating Expenses	
Warehouse Cost	150,000 SAR
Sales Team Salaries	200,000 SAR
Marketing Spend	100,000 SAR
Other Operating Expenses	50,000 SAR
Total Operating Expenses	500,000 SAR
EBITDA	
EBITDA	(115,000) SAR
EBITDA Margin %	(7.7%)
Depreciation, EBIT & Net Profit	
Monthly Depreciation (straight-line)	8,333 SAR
EBIT	(123,333) SAR
Corporate Tax (only if EBIT > 0)	- SAR
Net Profit	(123,333) SAR
Net Margin %	(8.3%)
Note: tax is modeled only on positive EBIT (no cash refund on losses). If EBIT is negative, this month generates a tax loss that could offset future taxable profits.	

Annual Cash Flow, ROI & IRR (5-Year Horizon)						
	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Annual Net Profit	- SAR	(1,480,000) SAR	(1,480,000) SAR	(1,480,000) SAR	(1,480,000) SAR	(1,480,000) SAR
Add back: Depreciation (non-cash)	- SAR	100,000 SAR	100,000 SAR	100,000 SAR	100,000 SAR	100,000 SAR
Operating Cash Flow	- SAR	(1,380,000) SAR	(1,380,000) SAR	(1,380,000) SAR	(1,380,000) SAR	(1,380,000) SAR
Initial Investment (WC + CapEx)	(2,500,000) SAR	- SAR	- SAR	- SAR	- SAR	- SAR
Working Capital Recovery (Year 5)	- SAR	- SAR	- SAR	- SAR	- SAR	2,000,000 SAR
Net Cash Flow	(2,500,000) SAR	(1,380,000) SAR	(1,380,000) SAR	(1,380,000) SAR	(1,380,000) SAR	620,000 SAR
Cumulative Cash Flow	(2,500,000) SAR	(3,880,000) SAR	(5,260,000) SAR	(6,640,000) SAR	(8,020,000) SAR	(7,400,000) SAR
Investment Metrics						
NPV @ Discount Rate	(6,489,443) SAR					
IRR (5-Year)	N/A — cash flows never turn positive					
Simple ROI (cumulative CF over horizon ÷ initial investment)	(296.0%)					
Payback Period (years, approx.)	Not recovered within horizon					
All years use the same steady-state monthly result from the 'Monthly P&L' tab (volume, pricing and costs are assumed flat across the 5-year horizon per the case data). Change any Assumptions cell to re-run every metric on this sheet.						

Break-Even Analysis															
Contribution per Unit Purchased															
Effective Revenue per Unit Purchased (net of waste)	1.49 SAR														
Variable Cost per Unit Purchased (purchase + logistics)	1.10 SAR														
Contribution Margin per Unit Purchased	0.39 SAR														
Fixed Costs (Monthly)															
Operating Expenses (Warehouse+Salaries+Marketing+O	500,000 SAR														
Depreciation	8,333 SAR														
Total Fixed Costs (incl. depreciation)	508,333 SAR														
Break-Even Volume															
Break-Even Volume — cash operating costs only (cans/m	1,298,701														
Break-Even Volume — incl. depreciation (cans/month	1,320,346														
Current Planned Volume (cans/month)	1,000,000														
Volume Surplus / (Deficit) vs. Break-Even	(320,346)														
Current Volume as % of Break-Even	75.7%														
Chart Data — Revenue vs. Total Cost by Volume															
Volume (cans/month)	Total Revenue	Total Cost													
-	- SAR	508,333 SAR													
250,000	371,250 SAR	783,333 SAR													
500,000	742,500 SAR	1,058,333 SAR													
750,000	1,113,750 SAR	1,333,333 SAR													
1,000,000	1,485,000 SAR	1,608,333 SAR													
1,250,000	1,856,250 SAR	1,883,333 SAR													
1,500,000	2,227,500 SAR	2,158,333 SAR													
1,750,000	2,598,750 SAR	2,433,333 SAR													
2,000,000	2,970,000 SAR	2,708,333 SAR													
Total Cost line includes fixed costs (opex + depreciation) plus variable cost per can (purchase + logistics). The break-even point is where the two lines cross.															

Break-Even Chart: Revenue vs. Total Cost



Sensitivity Analysis (Monthly EBITDA / Net Profit Impact)

Table 1 — Sales Volume Sensitivity (Price & Costs Held Constant)

Volume Change	(20.0%)	(10.0%)	-	10.0%	20.0%
Units Purchased (cans)	800,000	900,000	1,000,000	1,100,000	1,200,000
Units Sold (cans)	792,000	891,000	990,000	1,089,000	1,188,000
Revenue	1,188,000 SAR	1,336,500 SAR	1,485,000 SAR	1,633,500 SAR	1,782,000 SAR
COGS	800,000 SAR	900,000 SAR	1,000,000 SAR	1,100,000 SAR	1,200,000 SAR
Logistics Cost	80,000 SAR	90,000 SAR	100,000 SAR	110,000 SAR	120,000 SAR
Total Operating Expenses	500,000 SAR	500,000 SAR	500,000 SAR	500,000 SAR	500,000 SAR
EBITDA	(192,000) SAR	(153,500) SAR	(115,000) SAR	(76,500) SAR	(38,000) SAR
EBITDA Margin %	(16.2%)	(11.5%)	(7.7%)	(4.7%)	(2.1%)

Table 2 — Selling Price Sensitivity (Volume & Costs Held Constant)

Price Change	(10.0%)	(5.0%)	-	5.0%	10.0%
Selling Price (SAR/can)	1.35	1.43	1.50	1.58	1.65
Revenue	1,336,500 SAR	1,410,750 SAR	1,485,000 SAR	1,559,250 SAR	1,633,500 SAR
COGS + Logistics	1,100,000 SAR	1,100,000 SAR	1,100,000 SAR	1,100,000 SAR	1,100,000 SAR
Total Operating Expenses	500,000 SAR	500,000 SAR	500,000 SAR	500,000 SAR	500,000 SAR
EBITDA	(263,500) SAR	(189,250) SAR	(115,000) SAR	(40,750) SAR	33,500 SAR
EBITDA Margin %	(19.7%)	(13.4%)	(7.7%)	(2.6%)	2.1%

Table 3 — Monthly EBITDA Matrix: Volume (rows) x Selling Price (columns)

Volume ↓ / Price →	1.35	1.43	1.50	1.58	1.65
800,000	(310,800) SAR	(251,400) SAR	(192,000) SAR	(132,600) SAR	(73,200) SAR
900,000	(287,150) SAR	(220,325) SAR	(153,500) SAR	(86,675) SAR	(19,850) SAR
1,000,000	(263,500) SAR	(189,250) SAR	(115,000) SAR	(40,750) SAR	33,500 SAR
1,100,000	(239,850) SAR	(158,175) SAR	(76,500) SAR	5,175 SAR	86,850 SAR
1,200,000	(216,200) SAR	(127,100) SAR	(38,000) SAR	51,100 SAR	140,200 SAR

Table 3 shows monthly EBITDA (SAR) for combinations of volume and price shocks. Fixed operating expenses are held constant at the base-case level in all scenarios.

Analysis & Recommendations	
•	What drives profitability
	The per-can contribution margin (selling price – purchase price – logistics, adjusted for waste) is only 0.385 SAR — a thin 25.7% of the selling price. At this margin, profitability is extremely sensitive to volume, price, and cost control, since the 500,000 SAR/month fixed cost base (warehouse, salaries, marketing, other) must be covered before any profit is earned.
•	Base case is loss-making
	At the stated 1,000,000 cans/month, the model shows Revenue of 1,485,000 SAR against COGS + logistics of 1,100,000 SAR and fixed opex of 500,000 SAR, producing a negative EBITDA of roughly (115,000) SAR per month — before depreciation, tax, or financing.
•	Break-even is above planned volume
	Break-even volume (including depreciation) is about 1,320,000 cans/month — roughly 32% higher than the 1,000,000 cans/month assumed in the case. The contract as priced does not cover its own fixed costs at the expected volume.
•	Key risk: thin unit economics
	With only ~0.385 SAR contribution per can, a small adverse move in purchase price, selling price, or logistics cost can flip the business from marginal loss to deep loss, or from breakeven to solidly profitable — there is very little cushion.
•	Key risk: volume risk
	Retail sell-through, seasonality, and competitor activity (Coca-Cola, local brands) can easily move volume ±10-20% month to month; the sensitivity tables show this alone can swing monthly EBITDA by roughly 100,000+ SAR in either direction.
•	Key risk: working capital and cash
	2,000,000 SAR of working capital plus 500,000 SAR of CapEx (2,500,000 SAR total) is committed up front, while the operating business is currently cash-negative — extending the cash recovery period and raising liquidity risk if losses persist.
•	Sensitivity — volume down 10%
	A 10% volume decline compounds the existing shortfall, pushing monthly EBITDA further into negative territory (see Sensitivity tab, Table 1) — there is no volume cushion in the current pricing structure.
•	Sensitivity — price up 5%
	Because fixed costs do not move with price, even a modest 5% selling-price increase (to about 1.575 SAR/can) meaningfully narrows or closes the EBITDA gap at the same volume (see Sensitivity tab, Table 2) — pricing is the single highest-leverage lever.
•	Is this a good investment as structured?
	No — as priced, the contract does not generate positive operating cash flow in any projected year, so NPV is negative and IRR is undefined (cash flows never turn positive). The 2,500,000 SAR investment would not be recovered under the base-case assumptions.
•	What to tell the decision-maker
	Do not proceed at the current price and cost structure. Before committing capital, renegotiate the purchase price and/or retail price to lift the per-can margin, and/or right-size fixed overhead (warehouse, sales headcount, marketing) to the 1,000,000 cans/month volume; re-test break-even and IRR once revised terms are in hand.